

Rogers-Ramanujan Derivation Final Report

2026-05-15

Contents

Rogers-Ramanujan Derivation Final Report	1
Abstract	1
Introduction	1
Goal and Formal Setting	2
Proof Strategy at a Glance	3
Foundational Results and Diagnostics	3
Routes Explored Before the Transfer Mechanism	5
The Bailey-Style Triangular Transfer	6
Product-Side Closure	8
Final Proof Chain	9
Validation and Artifact Trail	10
Conclusions and Future Work	11
Conclusions	11
References	12

Rogers-Ramanujan Derivation Final Report

Abstract

This report synthesizes a completed formal-power-series derivation of the two Rogers-Ramanujan identities

$$\sum_{n \geq 0} \frac{q^{n^2}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+1})(1 - q^{5m+4})}$$

and

$$\sum_{n \geq 0} \frac{q^{n^2+n}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+2})(1 - q^{5m+3})}.$$

The proof works in $\mathbb{Z}[[q]]$, where equality is coefficientwise equality and infinite sums and products are interpreted by coefficientwise stabilization. The completed route derives finite q-binomial and Jacobi product identities, derives a Bailey-style triangular transfer from finite matrix inversion, applies that transfer to the two Rogers-Ramanujan series, and closes the product side by formal cancellation of modulo-5 Euler factors. The final audit status is “5 validated - promise_check=green”: milestones M0 through M4 are all validated with high confidence, with zero critical, moderate, or minor findings.

Introduction

The goal was to build a derivational proof rather than quote a known proof or rely on coefficient checks. The work therefore separates exploratory computation from proof-bearing algebra. Computations were used to test finite recurrences, reject false nearby identities, diagnose failed mechanisms, and guard against transcription errors. The

proof itself rests on formal lemmas: finite q-binomial algebra, a Jacobi product identity, a derived triangular matrix transform, coefficientwise limiting, and formal product cancellation in $\mathbb{Z}[[q]]$.

The report is organized as follows. It first states the formal setting and the accepted proof strategy. It then records the foundational diagnostics and the explored routes that did not close the proof. The central sections present the Bailey-style triangular transfer, product-side closure, and final proof chain. The last sections summarize validation artifacts, conclusions, residual bookkeeping debt, and references.

Goal and Formal Setting

The project goal was to derive the two Rogers-Ramanujan identities from first principles in a formal power series setting. The two identities are

$$\sum_{n \geq 0} \frac{q^{n^2}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+1})(1 - q^{5m+4})},$$

and

$$\sum_{n \geq 0} \frac{q^{n^2+n}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+2})(1 - q^{5m+3})}.$$

The work was carried out in $\mathbb{Z}[[q]]$, the ring of formal power series with integer coefficients. In this setting, equality means coefficientwise equality: two series are equal if the coefficient of each power q^k agrees. This avoids relying on analytic convergence for the proof. Infinite sums, infinite products, and limits are interpreted by coefficientwise stabilization: for any fixed degree k , only finitely many summands or product factors can affect the coefficient of q^k , so that coefficient eventually becomes fixed.

The finite q-Pochhammer symbol is

$$(q; q)_n = \prod_{k=1}^n (1 - q^k), \quad (q; q)_0 = 1.$$

More generally,

$$(a; q)_n = \prod_{i=0}^{n-1} (1 - aq^i), \quad (a; q)_\infty = \prod_{i \geq 0} (1 - aq^i),$$

when the infinite product is coefficientwise-defined. A formal power series with constant term 1 is a unit in $\mathbb{Z}[[q]]$, so the final product-side proof cancels such factors algebraically, not analytically.

The two series sides were denoted in the reports in two equivalent indexing conventions. In this final synthesis, write

$$S_0(q) = \sum_{n \geq 0} \frac{q^{n^2}}{(q; q)_n}, \quad S_1(q) = \sum_{n \geq 0} \frac{q^{n^2+n}}{(q; q)_n}.$$

The finite setup also gave a partition interpretation used during exploration. The summand $q^{n^2}/(q; q)_n$ generates partitions into exactly n positive parts with adjacent gaps at least two, by subtracting the staircase $(2n-1, 2n-3, \dots, 1)$. The summand $q^{n^2+n}/(q; q)_n$ generates the analogous partitions with smallest part at least two, by subtracting $(2n, 2n-2, \dots, 2)$. This interpretation supported the diagnostic partition and bijection work, but the completed proof ultimately used an algebraic transfer mechanism.

Source basis: `reports/cycles/report_cycles_1-3.md`, `reports/cycles/report_cycles_22-24.md`, `reports/cycles/report_cycles_25-26.md`, and the final audit summary for milestone M0.

Proof Strategy at a Glance

The accepted proof route was algebraic. It did not proceed by treating coefficient checks as proof, and it did not import an external Rogers-Ramanujan proof. The route that closed the derivation had four main steps.

First, the work established formal-power-series rules, finite truncation semantics, and reproducible finite experiments. Those experiments checked the target identities and nearby negative controls, but they were classified as discovery and regression support rather than theorem proofs.

Second, finite q -binomial algebra was derived internally. From that finite algebra, the run derived a Jacobi-type product identity:

$$(z; Q)_\infty (Q/z; Q)_\infty (Q; Q)_\infty = \sum_{j \in \mathbb{Z}} (-1)^j z^j Q^{j(j-1)/2}.$$

The two specializations needed later are obtained by taking $(Q, z) = (q^5, q^2)$ and $(Q, z) = (q^5, q)$. They produce the bilateral sums with exponents $j(5j-1)/2$ and $j(5j-3)/2$, respectively.

Third, the missing bridge from the Rogers-Ramanujan series to those Jacobi specializations was supplied by a derived triangular Bailey-style transform. The reports describe this as “Bailey-style” because it uses a lower triangular matrix relation and its inverse, but the needed special case was derived directly from finite q -binomial cancellations. This transform converted S_0 and S_1 into the corresponding bilateral Jacobi sums after multiplication by the Euler product $(q; q)_\infty$.

Fourth, product-side formalization converted the transformed identities into the target reciprocal residue-class products. The key factorization is

$$(q; q)_\infty = (q, q^2, q^3, q^4, q^5; q^5)_\infty.$$

After the triangular transform and Jacobi identity give

$$(q; q)_\infty S_0(q) = (q^2, q^3, q^5; q^5)_\infty$$

and

$$(q; q)_\infty S_1(q) = (q, q^4, q^5; q^5)_\infty,$$

formal cancellation of unit factors leaves exactly

$$S_0(q) = \frac{1}{(q, q^4; q^5)_\infty}, \quad S_1(q) = \frac{1}{(q^2, q^3; q^5)_\infty}.$$

The final audit summary records five validated milestones: M0 for the formal setup, M1 for finite approximant exploration, M2 for the transfer mechanism, M3 for product-side cancellation, and M4 for final proof synthesis and validation separation. Each milestone is recorded as validated with high confidence, and the final audit headline is “5 validated - promise_check=green.”

Source basis: `reports/cycles/report_cycles_7-9.md`, `reports/cycles/report_cycles_10-12.md`, `reports/cycles/report_cycles_13-14.md`, and the final audit summary.

Foundational Results and Diagnostics

The first substantive phase built the formal and computational foundation. The finite coefficient harness compared truncated series and product expansions through degree 40. The two true target pairs had no nonzero coefficient difference through that range. Negative controls failed early: residue classes $\{1, 3\}$ and $\{2, 4\}$ failed at degree 3, and the shifted exponent $n^2 + 2n$ failed at degree 1. These checks supported the target selection and caught false nearby identities, but they were not promoted to proof.

The first structural proof was a q -difference ladder for

$$F(z, q) = \sum_{n \geq 0} \frac{z^n q^{n^2}}{(q; q)_n}.$$

The proved identity was

$$F(z, q) - F(zq, q) = zqF(zq^2, q).$$

Specializing $z = q^r$ gives

$$A_r = A_{r+1} + q^{r+1}A_{r+2}, \quad A_r = F(q^r, q).$$

Here $A_0 = S_0$ and $A_1 = S_1$. The reports also record a tail-normalized uniqueness result: there is at most one solution of this ladder satisfying $A_r \rightarrow 1$ coefficientwise as $r \rightarrow \infty$. This proved useful series-side structure, but the product-side closure for the ladder was not yet established at that stage.

The same phase proved a finite largest-part recurrence for gap-two partitions. If $G_N^{(a)}$ generates gap-two partitions with all parts at least a and largest part at most N , then

$$G_N^{(a)} = G_{N-1}^{(a)} + q^N G_{N-2}^{(a)}.$$

This recurrence comes from splitting partitions according to whether the largest allowed part N appears. If it appears, the gap condition excludes $N-1$, leaving a residual partition counted by $G_{N-2}^{(a)}$.

The product-transform infrastructure began by multiplying by the Euler product

$$E(q) = (q; q)_\infty.$$

The target identities become equivalent to complementary modulo-5 product identities:

$$E(q)S_0(q) = (q^2; q^5)_\infty (q^3; q^5)_\infty (q^5; q^5)_\infty,$$

$$E(q)S_1(q) = (q; q^5)_\infty (q^4; q^5)_\infty (q^5; q^5)_\infty.$$

The finite q -binomial theorem and the Jacobi-type product identity supplied the product/Jacobi half of the proof. What remained open after this foundation was the transformed-series collapse: proving that $E(q)S_0$ and $E(q)S_1$ equal the corresponding bilateral Jacobi sums.

The finite transformed series used to study this bottleneck was

$$H_{\alpha, N}(q) = \sum_{n=0}^N q^{n^2 + \alpha n} \frac{(q; q)_N}{(q; q)_n}, \quad \alpha \in \{0, 1\}.$$

It satisfies the recurrence

$$H_{\alpha, N} = (1 - q^N)H_{\alpha, N-1} + q^{N^2 + \alpha N}.$$

The experiments showed stabilization against the natural bilateral targets, while simple Gaussian-window finite identities failed at degree 1. This result narrowed the open problem: the Jacobi product side was available, but the proof still needed a valid series-to-Jacobi transfer.

Source basis: `reports/cycles/report_cycles_1-3.md` and `MANIFEST.md` cross-references for `scripts/rr/finite_rr_experiments/rr/q_difference_probe.wls`, `docs/proof/finite_gap_recurrence.md`, and `scripts/rr/product_transform_prob`

Routes Explored Before the Transfer Mechanism

Several routes were explored before the triangular transform closed the proof. These routes are part of the report because they show what was derived, what was rejected, and why the final strategy changed.

The signed-object route expanded the quotient in $H_{\alpha,N}$ as

$$\frac{(q;q)_N}{(q;q)_n} = \prod_{s=n+1}^N (1 - q^s).$$

Each term became a signed object (n, S, α, N) , where

$$S \subseteq \{n+1, \dots, N\}, \quad w_\alpha(n, S) = n^2 + \alpha n + \sum_{s \in S} s, \quad \text{sgn}(n, S) = (-1)^{|S|}.$$

This expansion was proved and recorded as a valid representation of the transformed finite series. A local absorb/release rule was then tested as a sign-reversing cancellation mechanism. The accepted local moves were weight-preserving, sign-reversing, and involutive when defined, but the rule was not total. At $N = 12$, it left 1,887 of 4,059 tested $\alpha = 0$ objects unpaired and all 3,865 tested $\alpha = 1$ objects unpaired. The local rule was therefore rejected as a proof mechanism.

The nonlocal subset-transfer branch broadened the same idea. An upward move $n \mapsto n + r$ removed an odd subset $A \subseteq S$ satisfying

$$\sum A = r(2n + \alpha + r),$$

and a downward move $n \mapsto n - r$ added an odd missing subset satisfying

$$\sum A = r(2n + \alpha - r).$$

The branch found no weight, sign, or tail-condition failures for accepted transfers, but it proved structural obstructions. Singleton-tail objects created stable non-pentagonal no-move cases, such as $(0, \{5\})$ for $\alpha = 0$ and $(0, \{3\})$ for $\alpha = 1$. The canonical smallest- r variant also failed to be involutive. This rejected the tested pure nonlocal subset-transfer family.

The shifted-state recurrence branch started from the exact scalar recurrence for $H_{\alpha,N}$. Writing

$$H_{\alpha,N}(q) = \sum_{k \geq 0} h_{\alpha,N}(k) q^k,$$

it derived

$$h_{\alpha,N}(k) = h_{\alpha,N-1}(k) - h_{\alpha,N-1}(k - N) + \mathbf{1}_{k=N^2+\alpha N}.$$

This proves coefficient stabilization: for $N > k$, the coefficient $h_{\alpha,N}(k)$ no longer changes. For diagonal states

$$T_{\alpha,N}(d) = h_{\alpha,N}(N + d),$$

the branch proved

$$T_{\alpha,N}(d) = T_{\alpha,N-1}(d + 1) - h_{\alpha,N-1}(d) + \mathbf{1}_{d=N^2+(\alpha-1)N}.$$

The same formula explains why a fixed finite diagonal window does not close under modulo-5 splitting: the recurrence always requires an outside boundary state. This rejected the direct fixed-window modulo-5 recurrence approach.

The Euler-tail telescoping route multiplied the series by $(q; q)_\infty$ and expanded the tail by the finite q-binomial theorem. It proved the double-sum identity

$$(q; q)_\infty S_\alpha(q) = \sum_{n, k \geq 0} \frac{(-1)^k q^{n^2 + \alpha n + kn + k(k+1)/2}}{(q; q)_k}.$$

The tested low-order telescoping certificates had no solution, so this branch produced a useful expansion but not the needed collapse proof. The report explicitly limits that rejection to the tested ansatz class.

The direct partition-bijection route generated finite gap-two partition sets

$$G_\alpha(K) = \{\lambda \vdash K : \lambda_i - \lambda_{i+1} \geq 2, \lambda_{\ell(\lambda)} \geq 1 + \alpha\}$$

and residue-class partition sets

$$R_0(K) = \{\rho \vdash K : \rho_i \equiv 1, 4 \pmod{5}\}, \quad R_1(K) = \{\rho \vdash K : \rho_i \equiv 2, 3 \pmod{5}\}.$$

The exact finite enumerations matched in cardinality through $K_{\max} = 28$, but simple static signatures using length, beta-set runner counts, shifted beta-set runner counts, and quotient sums failed. An independent nearest-runner bead slide also failed because it changed total weight in minimal examples. The result was not a rejection of all direct bijections; it only showed that the tested static beta/abacus mechanisms were insufficient.

These explored routes narrowed the proof search. They supplied valid lemmas, regression data, and precise obstructions, but none gave the missing transformed-series collapse. The next successful step changed mechanism: instead of searching for a direct cancellation or bijection, the proof derived a triangular matrix transform that connected the two Rogers-Ramanujan series to the already-derived Jacobi products.

Source basis: `reports/cycles/report_cycles_4-6.md`, `reports/cycles/report_cycles_7-9.md`, and `MANIFEST.md` cross-references for `scripts/rr/transformed_cancellation_probe.wls`, `scripts/rr/nonlocal_involution_probe.wls`, `scripts/rr/mod5_state_recurrence_probe.py`, `scripts/rr/euler_tail_telescoping_probe.wls`, and `scripts/rr/direct_bijection_probe.py`.

The Bailey-Style Triangular Transfer

The proof-bearing bridge was the internally derived Bailey-style triangular transfer. Its role was to convert the two Rogers-Ramanujan series into the bilateral Jacobi sums already obtained from finite q-binomial algebra. This section uses the notation of the final report, where S_0 denotes the series with exponent n^2 and S_1 denotes the series with exponent $n^2 + n$.

The transfer begins with a finite lower triangular matrix. For a parameter a , define

$$M_{n,r}^{(a)} = \frac{1}{(q; q)_{n-r} (aq; q)_{n+r}}, \quad 0 \leq r \leq n.$$

The matrix relation is

$$\beta_n = \sum_{r=0}^n M_{n,r}^{(a)} \alpha_r.$$

The derived inverse, recorded as Lemma L33 in the source reports, is

$$\alpha_n = \sum_{j=0}^n \frac{1 - aq^{2n}}{1 - a} \frac{(a; q)_{n+j}}{(q; q)_{n-j}} (-1)^{n-j} q^{(n-j)(n-j-1)/2} \beta_j.$$

For $a = 1$, the report treats this formula through the specialized limiting case used in the proof, avoiding a literal 0/0 substitution. The derivation multiplied the finite triangular matrices directly. Diagonal terms reduced to 1, and off-diagonal terms reduced to finite q-binomial cancellations of the form

$$\sum_k (-1)^k q^{k(k-1)/2} \begin{bmatrix} m \\ k \end{bmatrix}_q q^{-tk} = 0, \quad 0 \leq t < m.$$

Applying the inverse to the common sequence

$$\beta_n = \frac{1}{(q; q)_n}$$

gave the two explicit alpha sequences needed for the Rogers-Ramanujan series. For $a = 1$,

$$\alpha_0^{(0)} = 1, \quad \alpha_r^{(0)} = (-1)^r q^{r(3r-1)/2} (1 + q^r) \quad (r \geq 1).$$

For $a = q$,

$$\alpha_r^{(1)} = (-1)^r q^{r(3r+1)/2} \frac{1 - q^{2r+1}}{1 - q}.$$

The coefficientwise limiting transform is

$$\sum_{n \geq 0} a^n q^{n^2} \beta_n = \frac{1}{(aq; q)_\infty} \sum_{r \geq 0} a^r q^{r^2} \alpha_r.$$

Its formal proof reduced the remaining inner sum to

$$\sum_{s \geq 0} \frac{z^s q^{s(s-1)/2} (zq^s; q)_\infty}{(q; q)_s} = 1.$$

The coefficient of z^m vanishes for every $m > 0$ because the finite q-binomial factor $(1; q)_m$ is zero. Coefficientwise interchange was justified by the earlier truncation lemmas, so the limiting transform is a formal-power-series statement rather than an analytic convergence claim.

With the explicit alpha sequences inserted, the transform identifies the two Rogers-Ramanujan series with bilateral Jacobi sums after multiplication by the appropriate Euler factor. For $a = 1$,

$$S_0(q) = \frac{1}{(q; q)_\infty} \sum_{j \in \mathbb{Z}} (-1)^j q^{j(5j-1)/2}.$$

For $a = q$, using $(q; q)_\infty = (1 - q)(q^2; q)_\infty$, the transformed equation becomes

$$S_1(q) = \frac{1}{(q; q)_\infty} \sum_{j \in \mathbb{Z}} (-1)^j q^{j(5j-3)/2}.$$

The earlier Jacobi product identity then turns these bilateral sums into the complementary modulo-5 products:

$$\sum_{j \in \mathbb{Z}} (-1)^j q^{j(5j-1)/2} = (q^2, q^3, q^5; q^5)_\infty,$$

and

$$\sum_{j \in \mathbb{Z}} (-1)^j q^{j(5j-3)/2} = (q, q^4, q^5; q^5)_\infty.$$

This transfer closed the transformed-series bottleneck. The source reports record milestone M2 as validated with high confidence, and `scripts/rr/bailey_matrix_probe.wls` is described as a regression check for the alpha solves,

triangular residuals, pair residuals, and limiting inner-sum residuals. The proof itself rests on the finite triangular inversion and limiting algebra, not on the residual check.

Source basis: `reports/cycles/report_cycles_7-9.md`, `MANIFEST.md` cross-references for `scripts/rr/bailey_matrix_probe.w` and `docs/proof/bailey_matrix_transform.md`, and the final audit summary for milestone M2.

Product-Side Closure

Once the triangular transfer supplied the two transformed identities, the remaining work was formal product-side cancellation in $\mathbb{Z}[[q]]$. Every product involved has constant term 1, so each product is a formal unit and may be canceled by multiplying by its inverse in the formal power series ring.

The Euler product factors by residue classes modulo 5:

$$(q; q)_\infty = (q, q^2, q^3, q^4, q^5; q^5)_\infty.$$

For the first Rogers-Ramanujan series, the transfer and Jacobi specialization give

$$(q; q)_\infty S_0(q) = (q^2, q^3, q^5; q^5)_\infty.$$

Substituting the Euler residue factorization and canceling unit factors gives

$$S_0(q) = \frac{(q^2, q^3, q^5; q^5)_\infty}{(q, q^2, q^3, q^4, q^5; q^5)_\infty} = \frac{1}{(q, q^4; q^5)_\infty}.$$

This is the product over parts congruent to 1 or 4 modulo 5:

$$\frac{1}{(q, q^4; q^5)_\infty} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+1})(1 - q^{5m+4})}.$$

For the second Rogers-Ramanujan series, the transformed identity is

$$(q; q)_\infty S_1(q) = (q, q^4, q^5; q^5)_\infty.$$

The same residue factorization gives

$$S_1(q) = \frac{(q, q^4, q^5; q^5)_\infty}{(q, q^2, q^3, q^4, q^5; q^5)_\infty} = \frac{1}{(q^2, q^3; q^5)_\infty},$$

which is

$$\frac{1}{(q^2, q^3; q^5)_\infty} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+2})(1 - q^{5m+3})}.$$

The source reports record these two product-side statements as L39 and L40, validated under milestone M3. `scripts/rr/product_side_formal_check.wls` supported this stage by checking transformed, Jacobi, and final reciprocal-product residuals, including the constant-term boundary case. Those computations are recorded as regression checks; the proof step is the formal unit cancellation shown above.

Source basis: `reports/cycles/report_cycles_10-12.md`, `MANIFEST.md` cross-references for `scripts/rr/product_side_formal` and `docs/proof/product_side.md`, and the final audit summary for milestone M3.

Final Proof Chain

The final proof can be read as a single formal chain in $\mathbb{Z}[[q]]$.

1. Work in the formal power series ring $\mathbb{Z}[[q]]$. Equality is coefficientwise equality, infinite sums and products are interpreted by coefficientwise stabilization, and products with constant term 1 are formal units.
2. Derive the finite q-binomial theorem by the Gaussian Pascal recurrence. This supplies the finite algebra needed for both the Jacobi product identity and the later triangular matrix cancellations.
3. Use the finite q-binomial theorem to derive the Jacobi-type product identity

$$(z; Q)_\infty (Q/z; Q)_\infty (Q; Q)_\infty = \sum_{j \in \mathbb{Z}} (-1)^j z^j Q^{j(j-1)/2}.$$

4. Specialize the Jacobi identity at $(Q, z) = (q^5, q^2)$ and $(Q, z) = (q^5, q)$. This gives

$$B_0(q) = (q^2, q^3, q^5; q^5)_\infty = \sum_{j \in \mathbb{Z}} (-1)^j q^{j(5j-1)/2}$$

and

$$B_1(q) = (q, q^4, q^5; q^5)_\infty = \sum_{j \in \mathbb{Z}} (-1)^j q^{j(5j-3)/2}.$$

5. Derive the triangular matrix relation

$$\beta_n = \sum_{r=0}^n \frac{\alpha_r}{(q; q)_{n-r} (aq; q)_{n+r}}$$

and its finite inverse. The inverse proof uses finite matrix multiplication and q-binomial cancellations, not an external Bailey lemma citation.

6. Apply the inverse to $\beta_n = 1/(q; q)_n$. The $a = 1$ specialization gives

$$\alpha_0^{(0)} = 1, \quad \alpha_r^{(0)} = (-1)^r q^{r(3r-1)/2} (1 + q^r) \quad (r \geq 1),$$

and the $a = q$ specialization gives

$$\alpha_r^{(1)} = (-1)^r q^{r(3r+1)/2} \frac{1 - q^{2r+1}}{1 - q}.$$

7. Pass through the coefficientwise limiting transform

$$\sum_{n \geq 0} a^n q^{n^2} \beta_n = \frac{1}{(aq; q)_\infty} \sum_{r \geq 0} a^r q^{r^2} \alpha_r.$$

This turns S_0 and S_1 into the Jacobi sums B_0 and B_1 after multiplication by $(q; q)_\infty$:

$$(q; q)_\infty S_0(q) = B_0(q), \quad (q; q)_\infty S_1(q) = B_1(q).$$

8. Replace B_0 and B_1 by their Jacobi product forms:

$$(q; q)_\infty S_0(q) = (q^2, q^3, q^5; q^5)_\infty,$$

$$(q; q)_\infty S_1(q) = (q, q^4, q^5; q^5)_\infty.$$

9. Use

$$(q; q)_\infty = (q, q^2, q^3, q^4, q^5; q^5)_\infty$$

and cancel formal unit factors to obtain

$$S_0(q) = \frac{1}{(q, q^4; q^5)_\infty}, \quad S_1(q) = \frac{1}{(q^2, q^3; q^5)_\infty}.$$

In expanded product notation, these are exactly

$$\sum_{n \geq 0} \frac{q^{n^2}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+1})(1 - q^{5m+4})},$$

and

$$\sum_{n \geq 0} \frac{q^{n^2+n}}{(q; q)_n} = \prod_{m \geq 0} \frac{1}{(1 - q^{5m+2})(1 - q^{5m+3})}.$$

The reports record the final theorem entries as L41 and L42. The validation record for milestone M4 states that no equality in this final proof is justified solely by coefficient checks. The checks remain useful for discovery, transcription control, and regression testing, while the proof chain above rests on formal algebraic lemmas.

Source basis: `reports/cycles/report_cycles_10-12.md`, `docs/proof/final_proof.md` as referenced in `MANIFEST.md`, and the final audit summary for milestone M4.

Validation and Artifact Trail

The artifact record separates proof-bearing documents from exploratory and regression scripts. The workspace manifest lists 20 scripts under `scripts/rr`: 10 Wolfram scripts and 10 Python scripts, totaling 2,571 lines. It also lists 13 proof, lemma, and validation documents totaling 3,036 lines, and 82 data or figure files under `data/finite_experiments`.

The proof-bearing documents named in the final chain are `docs/proof/bailey_matrix_transform.md`, `docs/proof/product_side.md`, `docs/proof/final_proof.md`, and `docs/lemmas/lemma_catalogue.md`. The validation document, `docs/validation.md`, records which computations are discovery or regression checks and which results are actual proof steps.

The computational artifacts have distinct roles:

- `scripts/rr/finite_rr_experiments.wls` generated truncated coefficient comparisons and negative controls for discovery.
- `scripts/rr/q_difference_probe.wls` checked the series-side ladder residuals and related backsolve comparisons.
- `scripts/rr/product_transform_probe.wls` checked the finite transformed recurrence, finite Jacobi residuals, and failed simple Gaussian-window candidates.
- `scripts/rr/transformed_cancellation_probe.wls`, `scripts/rr/nonlocal_involution_probe.wls`, and `scripts/rr/mod5_state_recurrence_probe.py` supported rejected or partial transformed-series routes.
- `scripts/rr/euler_tail_telescoping_probe.wls` supported the Euler-tail expansion and rejected the tested low-order certificate ansatz.
- `scripts/rr/direct_bijection_probe.py` enumerated finite gap-two and residue partitions and rejected tested static beta/abacus mechanisms.
- `scripts/rr/bailey_matrix_probe.wls` checked the derived triangular transform, explicit pairs, and limiting inner-sum residuals.
- `scripts/rr/product_side_formal_check.wls` checked transformed, Jacobi, and final product residuals for product-side closure.

The source reports emphasize that these scripts did not replace proof. They made finite tests reproducible, caught negative controls, tested candidate mechanisms, and guarded against transcription errors after proof-bearing lemmas were written.

The figure record is similarly supportive. The final audit summary records 9 figures in the ledger and 9 figures present, with no missing or orphan figures. Those figures illustrate residual checks, support patterns, cancellation survivors, and rejected-route diagnostics; they are not additional assumptions in the theorem proof.

The final audit summary reports zero critical findings, zero moderate findings, and zero minor findings for the final run judgment. It also records no reconciliation events and a green promise check. The milestone distribution is 5 validated, with M0 through M4 all validated at high confidence.

Source basis: `MANIFEST.md`, `reports/cycles/report_cycles_7-9.md`, `reports/cycles/report_cycles_10-12.md`, later closure source reports, and the final audit summary.

Conclusions and Future Work

The mathematical directive is complete. The run produced a formal-power-series proof of both Rogers-Ramanujan identities, with the proof route resting on finite q -binomial algebra, Jacobi product specialization, an internally derived Bailey-style triangular transform, coefficientwise limiting, and formal modulo-5 product cancellation.

All five tracked milestones are validated with high confidence:

Milestone	Status	Role
M0	validated	Formal setup and truncation semantics
M1	validated	Finite approximant exploration and reproducible diagnostics
M2	validated	Bailey-style triangular transfer mechanism
M3	validated	Product-side formal cancellation
M4	validated	Final proof synthesis and validation separation

The final audit headline is “5 validated - promise_check=green.” The wall-cap flag is false.

The only residual debt recorded in the final audit summary is bookkeeping: the latest run-start ledger state remained marked as in-progress/high. The audit explicitly classifies this as a nonterminal ledger-state issue, not a proof defect for milestones M0-M4. The corresponding future work is to close or supersede that run-start bookkeeping entry after the final audit artifacts are committed.

An independent direct partition bijection remains a possible extension. It was not needed for the completed proof, and the explored static beta/abacus candidates were documented as insufficient. Pursuing a direct bijection would be a separate objective, not unfinished work required for the two identities proved here.

Source basis: final audit summary, `reports/cycles/report_cycles_40-42.md`, and later closure source reports.

Conclusions

The completed derivation proves both Rogers-Ramanujan identities in the formal power series ring $\mathbb{Z}[[q]]$. The proof is self-contained in the sense required by the run: each major equality is supported by finite algebraic lemmas, coefficientwise limiting arguments, or formal product cancellation, while computational artifacts are kept in a diagnostic and regression role.

Final validation status is “5 validated - promise_check=green.” Milestones M0 through M4 are all validated with high confidence; the final audit reports zero critical, moderate, or minor findings, no reconciliation events, and full figure coverage. The wall-cap flag is false. The only recorded residual debt is a bookkeeping item about closing or superseding a run-start ledger entry, not a mathematical proof defect.

References

No external references were cited in the final proof record. The report is based on the internal source reports, proof documents, validation document, manifest, and final audit summary listed throughout the body.